The Middle East Water Crisis

GERD, Desalination, and the World's Most Water-Stressed Region

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Executive Summary

The Middle East and North Africa region constitutes the world's most acutely water-scarce geography by every meaningful metric. Seventeen of the twenty-five most water-stressed countries globally are located within MENA boundaries , a concentration that reflects both the structural aridity of the Saharan and Arabian landmasses and decades of mismanaged extraction of finite hydrological reserves. Average water availability across the region stands below 1,000 cubic metres per person per year against a global average of approximately 6,000 m3, a disparity that frames an existential rather than merely environmental challenge .

The Arabian Peninsula's oil-wealthy states — Saudi Arabia, the UAE, Kuwait, and Qatar — possess effectively zero natural freshwater rivers or permanent lakes, rendering them structurally dependent on two mechanisms: industrial-scale desalination of seawater and extraction of fossil aquifers laid down tens of thousands of years before modern civilisation. Saudi Arabia derives more than 50% of its drinking water supply from desalination , while the UAE desalinates over 42% of its municipal supply . These engineering achievements disguise the underlying fragility of a system in which water production requires colossal and continuous energy inputs, and in which the geological reserves underpinning supplementary extraction are being consumed at rates orders of magnitude beyond natural recharge.

Climate change amplifies every dimension of this crisis. Mean temperatures across MENA are rising at approximately twice the global average rate, accelerating evaporation, reducing snowpack that feeds the region's few rivers, and driving human populations toward the very coastal and urban zones where water infrastructure is most stressed . The water crisis is simultaneously a food crisis, a migration driver, and a geopolitical flashpoint. The competition over the Nile between Egypt, Sudan, and Ethiopia — centred on the Grand Ethiopian Renaissance Dam — represents the sharpest illustration of how resource scarcity translates directly into security risk. The following analysis disaggregates the MENA water emergency by geography, technology, and politics, tracing the multiple vectors through which water scarcity is reshaping the regional order.

Part I — The MENA Water Deficit: Scale, Trajectory, and Climate Amplification

The fundamental hydrological condition of MENA is one of structural deficit. The region encompasses approximately 6% of the world's population while holding barely 1% of the world's renewable freshwater resources . This imbalance, already acute, is worsening on both axes simultaneously: population is growing while freshwater availability is declining. Per capita freshwater availability across MENA fell from approximately 3,430 cubic metres per year in 1960 to roughly 760 m3 per year by 2020, and with MENA's population projected to grow from 420 million toward 600 million or more by 2050, the per capita figure will continue its structural decline even absent any change in the physical supply of water .

The Intergovernmental Panel on Climate Change's Sixth Assessment Report delivers a particularly severe prognosis for MENA. Mean temperature across the region is rising at approximately twice the global average rate. Under the moderate RCP4.5 emissions scenario, many MENA subregions face precipitation declines of 10 to 25% by 2100, while heatwave frequency and intensity increase dramatically — outcomes that will simultaneously reduce inflows to rivers and reservoirs and increase agricultural and urban water demand . The Dead Sea water level has been falling by roughly one metre per year for decades, a visible symptom of the larger regional drawdown .

Individual country cases illustrate the range and severity of the crisis. Jordan, one of the world's most water-impoverished states, has renewable per capita freshwater availability of only approximately 89 cubic metres per person per year, placing it among the most acutely water-stressed nations on earth . Yemen's Sanaa Basin, home to the capital city and several million people, has experienced water table drops of between 2 and 6 metres per year in recent decades, a rate of depletion that places the aquifer on a trajectory toward exhaustion within a generation — a threat compounded by the total collapse of water infrastructure management during the civil war .

Egypt presents a different structural vulnerability: near-total dependence on a single external river. Approximately 97% of Egypt's freshwater derives from the Nile, and Egypt's treaty allocation under the 1959 Nile Waters Agreement is fixed at 55.5 billion cubic metres per year . With Egypt's population approaching 110 million and growing, this fixed allocation translates into a per capita share that diminishes every year. Libya's Great Man-Made River — a $25 billion infrastructure project constructed between 1984 and 2010, consisting of 1,750 kilometres of pipeline conveying water from Saharan fossil aquifers to the Mediterranean coast — represents perhaps the most dramatic expression of MENA's hydrological predicament: a nationally strategic infrastructure drawing on a non-renewable geological resource at unsustainable rates .

Part II — Desalination: The Gulf's Water Solution and Its Limits

Desalination has enabled the Arabian Peninsula's rapid modernisation, sustaining populations in environments that would otherwise be uninhabitable at current densities. Saudi Arabia operates the world's largest national desalination capacity: approximately 5.4 million cubic metres per day across 30 major plants managed by the Saudi Water Partnership Company . This infrastructure represents a triumph of industrial engineering. Yet its economics expose the fundamental vulnerability of the Gulf water model. Desalinated water costs between $0.50 and $1.50 per cubic metre to produce, compared to approximately $0.10 per cubic metre for conventional groundwater extraction — a cost differential that would be prohibitive for agricultural use and that makes desalinated water economically viable only because Gulf states have historically subsidised water prices far below production cost .

The energy intensity of desalination represents the most critical systemic constraint. Thermal desalination technologies — multi-stage flash (MSF) and multi-effect distillation (MED), which predominate in older Gulf plants — consume between 10 and 15 kilowatt-hours per cubic metre of freshwater produced. Even modern seawater reverse osmosis (SWRO) technology requires 3 to 4 kWh per cubic metre . Scaling up desalination to serve projected MENA populations therefore demands massive and growing energy inputs, creating a structural linkage between water security and energy production that has profound climate implications — particularly given that Gulf desalination has historically been powered by co-generation with fossil fuel plants.

The renewable energy transition offers a partial solution. Saudi Arabia's NEOM megaproject has planned for 100% solar-powered desalination using a combination of 4 gigawatts of solar capacity with SWRO technology . The UAE's Masdar initiative and the Mohammed bin Rashid Al Maktoum Solar Park are being integrated with SWRO desalination facilities, while Dubai Electricity and Water Authority (DEWA) operates desalination at a capacity of 727 million imperial gallons per day . Qatar's Ras Abu Fontas complex and Kuwait's Az Zour North Independent Water and Power Project (2,500 MW plus 486 MIGD water) represent the continued expansion of Gulf desalination infrastructure .

Globally, total desalination capacity is projected to approximately double, from 95 million cubic metres per day in 2020 to a projected 190 million m3/day by 2030 . The MENA region will account for the largest share of this expansion. However, this scaling trajectory confronts an environmental constraint that has received insufficient policy attention: brine disposal. Every cubic metre of freshwater produced by reverse osmosis generates approximately 1.5 cubic metres of hypersaline brine concentrate, which is discharged back into the sea. The Persian Gulf's semi-enclosed geography, high natural salinity, and shallow depth make it particularly vulnerable to brine accumulation. Studies have documented measurable salinity increases in the near-shore zones surrounding major Gulf desalination plants, with consequent damage to marine ecosystems including coral reefs, seagrass beds, and fisheries that Gulf coastal communities depend upon .

Part III — The Nile Basin Dispute: Egypt, Ethiopia, and the GERD Crisis

The Grand Ethiopian Renaissance Dam (GERD) on the Blue Nile represents the most consequential water infrastructure project in Africa and the source of the region's most dangerous inter-state water dispute. The dam, with a reservoir capacity of 74 billion cubic metres — the largest in Africa — and a hydropower generation capacity of 6,000 megawatts, was constructed between 2011 and 2022 . Its first filling began in 2020, triggering immediate alarm in Cairo and Khartoum. The reservoir's capacity alone approximates 1.5 times Egypt's entire annual Nile allocation under the 1959 treaty, giving Ethiopia extraordinary leverage over Egypt's primary water source during any drought year when the Nile's natural flow is reduced and Ethiopia chooses to limit GERD releases.

Egypt's position is existential in the literal sense. The country derives approximately 97% of its freshwater from the Nile, and its entire modern agricultural system — irrigating the Nile Delta and the narrow river valley that constitutes Egypt's arable land — depends on maintaining adequate flow downstream of the GERD site. Egypt regards the 1959 Nile Waters Agreement, which allocated 55.5 billion cubic metres annually to Egypt and 18.5 billion to Sudan, as a binding international legal commitment . President al-Sisi has characterised GERD as an existential threat and has stated publicly on multiple occasions that Egypt will not allow its Nile waters to be reduced — with Egyptian officials explicitly raising the military option in 2013, when the then-president stated Egypt would "go to war" if necessary .

Ethiopia's position is equally coherent and legally defensible from a different framework. Ethiopia is home to approximately 120 million people, a significant proportion of whom lack reliable electricity access, and GERD's 6,000 MW hydropower capacity represents a transformative development resource . Ethiopian authorities invoke the principle of sovereign right over natural resources and note that upstream states were not party to the 1959 Nile Waters Agreement — an agreement negotiated between Egypt and Sudan without consultation with upstream riparian states. The African Union mediation process, initiated formally in 2020, failed to produce a binding agreement through multiple rounds of talks in 2021, 2022, and 2023 .

As of August 2026, the GERD is operational and generating power. The fourth filling of the reservoir was completed in 2023, meaning the dam has now accumulated a substantial water reserve. Egypt's military strike option has been widely assessed as strategically problematic: a military attack on GERD — a massive concrete structure approximately 1,800 kilometres from the Egyptian border via landlocked Ethiopian territory — presents enormous logistical challenges, would constitute an act of war against a fellow African Union member, and risks triggering a humanitarian catastrophe downstream if the dam is damaged . Sudan's position remains strategically ambiguous: Khartoum benefits from GERD's regulation of flood-season flows and has been cautious about aligning with Egyptian pressure. The Nile dispute remains unresolved, with no binding legal framework governing GERD operations and a structural conflict of interest between upstream development and downstream water security that is unlikely to be peacefully resolved without significant diplomatic innovation.

Part IV — Israel's Water Technology: Engineering Abundance in Scarcity

Israel's management of water scarcity represents the most studied example of technological adaptation to structural aridity, and the country's water sector innovations have been diffused globally. The foundational technology is drip irrigation, developed in the 1960s by Simcha Blass at Kibbutz Hatzerim in the Negev, commercialised by the company Netafim, which has since become a global agricultural technology leader . Israel's agricultural sector achieved a 50% reduction in water use between the 1970s and 2010s while tripling crop production — a productivity transformation that is analytically inseparable from the systematic adoption of drip irrigation, precision farming, and metered water pricing .

On the supply side, Israel has constructed five major seawater desalination plants that collectively provide approximately 80% of the country's municipal water supply: the Sorek 1 and Sorek 2 plants (among the world's largest SWRO facilities), and facilities at Ashkelon, Hadera, and Ashdod . The annual revenue from desalination operations amounts to approximately NIS 2.3 billion, equivalent to roughly $630 million . On the wastewater side, Israel recycles more than 90% of its treated wastewater for agricultural reuse, compared to a global average of approximately 2% — the highest rate of wastewater recycling in the world . The Dan Region Reclamation Project, which serves the Tel Aviv metropolitan area, is the world's largest wastewater recycling facility by treated volume dedicated to agricultural reuse .

Israeli water technology exports have grown into a substantial economic sector. Water technology companies based in Israel — including Aqwise, TaKaDu, Amiad Water Systems, and numerous others alongside the pioneering Netafim — collectively generate approximately $2.5 billion in annual exports . The Abraham Accords have created new commercial pathways for Israeli water technology into the Gulf. Mekorot, Israel's national water company, has entered partnership discussions with the Abu Dhabi Water and Electricity Company (ADWEC), and Israeli water technology firms have established operational presences in the UAE . The 1994 Israel-Jordan Peace Treaty committed to a water-sharing arrangement allocating 55 million cubic metres per year from the Jordan River to Jordan. A 2021 expansion deal — facilitated by UAE involvement in an electricity-for-water swap structure — added a further 200 million cubic metres per year of additional supply commitment, representing a significant enhancement of Jordan's water security through regional cooperation .

Part V — Saudi Arabia's Fossil Water Crisis: The Aquifer Reckoning

Saudi Arabia's water crisis is perhaps the starkest example of the unsustainable developmental trajectory that characterised the mid-twentieth century in arid regions. Between the 1970s and 1990s, Saudi Arabia achieved wheat self-sufficiency through an ambitious agricultural programme centred on centre-pivot irrigation of the Wajid Aquifer — a fossil water reserve of approximately 38,000 years of geological age, replenished at rates negligible by human planning timescales . By 1984, Saudi Arabia achieved full wheat self-sufficiency, and by the 1990s it had become the world's sixth-largest wheat exporter . This achievement was purchased at the cost of fossil water extraction that could not be sustained indefinitely.

The scale of the extraction was assessed by hydrological surveys as catastrophic for the long-term viability of the aquifer. Agricultural operations consumed approximately 2,200 cubic metres of water per hectare per crop cycle — drawn entirely from geological reserves with no meaningful natural recharge . The Saudi government recognised the unsustainability and took the consequential decision in 2008 to phase out domestic wheat production entirely by 2016, accepting permanent dependence on imported wheat as the price of aquifer preservation . Saudi Arabia now imports 100% of its wheat consumption, totalling more than 3.5 million tonnes annually, historically sourced primarily from Russia and Ukraine — a supply chain that the Red Sea disruption of 2023-2024 exposed as geopolitically vulnerable .

The agricultural drawdown has not fully ceased. Al-Safi Danone, operating the world's largest integrated dairy farm on Saudi soil with approximately 168,000 cows and producing 2.5 million litres of milk per day, sustains its herd through alfalfa cultivation in the Saudi desert — alfalfa being among the most water-intensive crops in existence . The Saudi government issued a 2023 policy directive to ban alfalfa cultivation domestically by 2030, recognising that the dairy sector's fossil water consumption represents a continuation of the very extraction pattern that the 2008 wheat decision sought to address . Groundwater depletion in the Riyadh basin is estimated at between 1.0 and 1.5 metres per year . The combination of desalination dependency, fossil water depletion, and food import vulnerability creates a multi-layered structural fragility in the country that holds the world's largest oil reserves — a civilisational irony of considerable geopolitical significance.

Conclusion

The MENA water crisis is best understood not as an environmental management failure but as an existential civilisational challenge that drives food import dependency, massive energy expenditures for desalination, and geopolitical conflicts over shared water resources. The Nile dispute between Egypt and Ethiopia, the competition over Euphrates-Tigris flows between Turkey, Syria, and Iraq, and the Jordan River's chronic over-extraction collectively illustrate that water is a primary cause of conflict in the region — one that climate change is actively intensifying.

The technological responses — Israeli innovation, Gulf desalination, Saudi agricultural reform — demonstrate that human ingenuity can partially offset structural scarcity. But they cannot eliminate it. Every desalination plant requires energy. Every fossil aquifer extraction is irreversible. Every degree of regional warming reduces precipitation and increases evaporative demand. The water emergency is not a future scenario; it is the present condition of 600 million people, worsening on the timescale of years, not decades. MENA water policy is therefore inseparable from energy policy, food security policy, and regional geopolitics — a convergence of pressures that will define the region's stability through the middle of this century.

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